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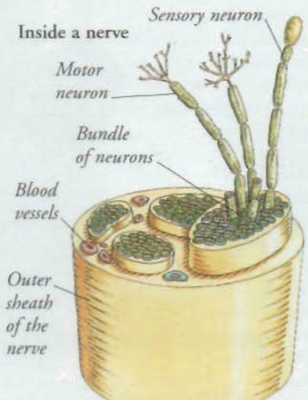
BRAIN AND NERVOUS SYSTEM

EVERY THOUGHT YOU HAVE, every emotion you feel, and every action you take is a reflection of the nervous system at work. At the core of the nervous system are the brain and spinal cord, known as the central nervous system (CNS). The most complex part of the CNS is the brain; this constantly receives information from the body, processes it, and sends out instructions telling the body what to do. The CNS communicates with every part of the body through an extensive network of nerves. The nerves and the CNS are both constructed from billions of nerve cells called neurons.

Nerves

Nerves form the "wiring" of the nervous system. Each nerve consists of a bundle of neurons (nerve cells) held together by a tough outer sheath. Nerves spread out from the brain and spinal cord and branch repeatedly to reach all parts of the body. Most nerves contain sensory neurons that carry nerve impulses towards the CNS, and motor neurons that carry nerve impulses away from the CNS.



Nerve endings
At the ends of sensory neurons there are nerve endings called sensory receptors. If you touch an object, a sensory receptor in the skin is stimulated, nerve impulses travel to the brain along the sensory neuron, and you feel the object. In this way, visually impaired people can "read" the Braille language with their fingertips.

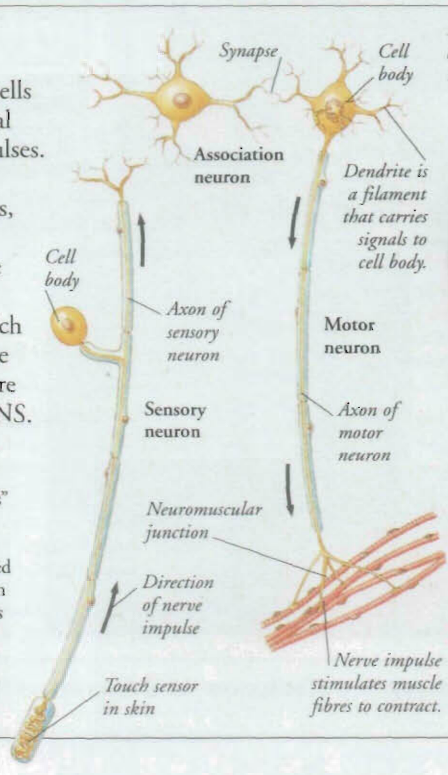
Nervous system

The nervous system is made up of the CNS and the peripheral nervous system, which consists of the nerves. The peripheral nervous system has two sections: the somatic system which controls voluntary actions, and the autonomic nervous system which controls automatic functions such as heart rate.

Neurons

Neurons are long, thin cells adapted to carry electrical signals called nerve impulses. There are three types of neurons: sensory neurons, motor neurons, and association neurons. The most numerous are association neurons, which transmit signals from one neuron to another and are found only inside the CNS.

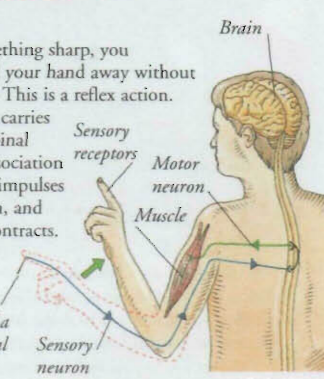
Nerve impulses
Nerve impulses are the "messages" that travel at high speed along neurons. Impulses are weak electrical signals that are generated and transmitted by neurons when they are stimulated. The stimulus may come from a sensory nerve ending, or from an adjacent neuron. Nerve impulses travel in one direction along the neuron.



Neuromuscular junction is a synapse between a motor neuron and muscle fibre.

Synapses
A synapse is a junction between two neurons. At a synapse, neurons do not touch. Instead, there is a tiny gap. When a nerve impulse reaches a synapse it triggers the release of chemicals, which travel across the gap and stimulate the second neuron to generate a nerve impulse.

Reflex actions
If you touch something sharp, you automatically pull your hand away without thinking about it. This is a reflex action. A sensory neuron carries impulses to the spinal cord, where an association neuron transmits impulses to a motor neuron, and the arm muscle contracts.



Santiago Ramón y Cajal

Spanish anatomist Santiago Ramón y Cajal (1852–1934) pioneered the study of the cells that make up the brain and nerves. He developed methods for staining nerve cells so they could be seen clearly under the microscope. His work revolutionized the examination of brain tissue.